Venkata Lakshmi Parimala Pasupuleti

AI Engineer · LLM Guardrails · Model Evaluation · Agentic AI Systems · Multi-Model Orchestration · Responsible AI · Python · Production

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AgentProbe (live eval toolkit) | McLean, Virginia · Open to Onsite

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production experience, specializing in **LLM guardrails, model evaluation, multi-agent orchestration, and responsible AI systems**. Currently at Intuit integrating Anthropic Claude (foundation model) into financial products -- building production guardrails, structured-output validation, LLM inference pipelines, and LLM-as-judge regression harnesses. Built **AgentProbe**: an open-source LLM evaluation toolkit with a **5-metric eval engine, multi-model orchestration across 5 providers, and cost-performance routing** -- 93.6% precision across 1,240 test cases. Built **Leasel**: an 8-agent LangGraph platform enforcing responsible AI in code -- LLM reasons, human or deterministic logic decides every regulated call; 100% citation correctness, 0% unsupported-claim rate, safety boundary never violated on adversarial tests. 7+ years at Entrata (100K+ MAU SaaS) building production Python/Java pipelines, async systems, and AWS infrastructure at scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

LLM Guardrails & Responsible AI | Model Evaluation & LLM-as-Judge | Multi-Model Orchestration | Agentic AI & Multi-Agent Systems | LangGraph & LangChain | RAG & Similarity Search | HITL & Model Governance | LLM Inference Pipelines | Observability & LLMOps | Python & FastAPI | AWS / Azure / GCP | Production AI at Scale

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — Foundation Model Integration · LLM Guardrails · Evaluation · Python · AWS

- Integrate **Anthropic Claude (foundation model)** into Python/FastAPI financial services -- design structured-output schemas and **LLM inference** pipelines for financial summaries and natural-language query answering; ground responses in customer data via **similarity search** and retrieval-augmented generation (RAG) for full citation traceability.
- Build **production guardrails** around Claude calls: input validation, sensitive-data redaction, output schema enforcement, and fallback logic when model output fails checks -- responsible AI enforced at the code layer, not the prompt layer.
- Build **LLM evaluation harness** with LLM-as-judge scoring and golden datasets for regression detection; CI-integrated via GitHub Actions -- caught 3 prompt regressions before release, zero hallucination incidents post-deployment; establish **model governance** standards for all Claude-powered APIs.
- Design **cost-performance optimization** for LLM inference: structured outputs reduce token waste; provider selection by latency/cost policy; **multi-model orchestration** across different financial query types. Containerize with Docker, ship to AWS with OpenTelemetry observability.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · AWS · PostgreSQL · Distributed Systems

- Shipped 20+ production features end-to-end (Python, Java/Spring Boot, PHP, PostgreSQL, React) across 120+ Agile sprints; reviewed 300+ PRs; mentored junior engineers on production standards and automated testing.
- Built **RabbitMQ async processing system** with idempotency guarantees for responsible bulk processing (gross-rent updates across thousands of regulated tenant records) -- designed retry logic eliminating double-processing failures; owned through production incidents and annual regulatory release cycles.
- Owned HUD income-limit ETL pipeline for 7 years: few thousand federal rows → 180K domain records with validation, rollback checks, and compliance enforcement -- zero violations across 7 annual release cycles; income limit errors affect resident housing eligibility, so correctness is non-negotiable.
- AWS S3 PHI migration with **Libsodium encryption** (100% PHI compliance, zero security incidents); optimized 200+ SQL queries (30% gain); Docker containerization cutting microservices execution time 50%.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Python · PyTorch · HuggingFace · Statistical Modeling

- Built predictive models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection; built fine-tuned classification model with HuggingFace + PyTorch (presented to faculty panel). Mentored 600+ students across Python and database design.

PORTFOLIO PROJECTS

Leasel — 8-Agent LangGraph Platform with Guardrails & HITL

Responsible AI | Agentic AI | Guardrails

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF + similarity search, SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Responsible AI enforced in code**: LLM extracts and reasons; a human or deterministic validator makes every final eligibility, compliance, and financial call -- no exceptions, even at high confidence. Tenant self-service agent safety boundary tested adversarially: never auto-resolves regulated decisions for a user.
- 8-agent LangGraph StateGraph with real conditional edges and **confidence-gated HITL escalation**: triage → retrieval → config/history/structured-data agents → reasoning → validation → human escalation. Correctly traces false-ineligibility ticket to misconfigured setting tagged to specific release.
- RAG with **similarity search** (hand-rolled TF-IDF + cosine reranking): 100% citation correctness, 0% unsupported-claim rate, 93.6% precision and 87.9% F1 across 1,240 test cases (validated by AgentProbe eval harness).

AgentProbe — LLM Evaluation Toolkit & Multi-Model Orchestration

Model Evaluation | LLM-as-Judge | Multi-Model Orchestration

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- Built **multi-model orchestration** system: LLM Router dynamically routes across 5 providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) based on **cost-performance policy** -- task classification selects provider by latency/cost/capability tradeoff; ~40% cost reduction vs always-routing to highest-capability model.
- **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression detection blocks production deploys when quality drops below threshold; 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) handle multi-step evaluation and auto-remediation. 100% local, zero vendor dependency.

AI on FHIR — Clinical NLP with Responsible AI & HIPAA Guardrails

Responsible AI | HIPAA Compliance

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP pipeline translating English/Spanish clinical questions into FHIR API calls using LLMs, BERT, and **similarity search** (embeddings); **HIPAA-compliant guardrails**: role-based access control for data isolation, output filtering for PHI; ~60% reduction in manual clinical lookup time.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

AI & LLMs: LLM Guardrails, Model Evaluation, LLM-as-Judge, LLM Inference, Multi-Model Orchestration, Foundation Model Integration, Model Governance, Agentic AI, Multi-Agent Systems, LangGraph, LangChain, HITL, Human-in-the-Loop, Responsible AI, Structured Outputs, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, Prompt Engineering, Hallucination Detection, Cost-Performance Optimization

RAG & Retrieval: RAG Pipelines, Retrieval-Augmented Generation, Similarity Search, Hybrid Search, Vector Database, ChromaDB, Pinecone, Weaviate, Elasticsearch, TF-IDF, Embeddings, Semantic Search, BERT, HuggingFace, spaCy, Document Intelligence, OCR

Observability & LLMOps: OpenTelemetry, Prometheus, Grafana, LLMOps, MLflow, CI/CD for LLM Systems, Regression Detection, Golden Datasets, Eval-Driven Development, Agent Observability, A/B Testing, GitHub Actions

Python & ML: Python (9+ yrs), FastAPI, Flask, Django, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark, Java, Spring Boot, TypeScript, Node.js, React, PHP, Kotlin, Scala, GraphQL, SQL, PL/SQL, Bash

Databases & Data: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization, Stored Procedures

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, RabbitMQ, Kafka, Celery